

A Computer Vision-Based Physical Activity Application for Children with Autism

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Abstract—Autism Spectrum Disorder (ASD) is a neurological condition characterized by challenges in social interaction, communication, repetitive behaviors, and restricted interests. While the exact cause remains unknown, ASD results from genetic and environmental factors and persists throughout life. Early diagnosis and intervention enhance quality of life and potential. Games have proven effective in nurturing socialization, skill acquisition, and adaptability in children with ASD, leveraging their structured and visually appealing design.

This study adopted a game-based approach to support motor skills, socialization, and learning in children with ASD. Using computer vision techniques, children's movements were captured via cameras and transformed into interactive virtual characters in real time. Activities included movement mimicry games to improve attention and motor skills, dance games for physical coordination, and turn-based games to enhance social skills. The system, developed in this study using Google MediaPipe Pose and Unity, is cost-effective and suitable for preschools and home use. Feedback from educators at the Autism Sports and Life Center guided system refinement. The work presented in this article offers a robust foundation for fostering independence and social participation in children with ASD.

Index Terms—Autism, Unity, Google MediaPipe, Spectrum Disorder, Game, Physical Activity, Socialization

I. INTRODUCTION

A. Originality

Play is a crucial medium through which children with Autism Spectrum Disorder (ASD) can experience their first real-life encounters. During play, they have the opportunity to learn and establish cause-and-effect relationships. Play facilitates the building of relationships with the child, helps uncover the child's lifestyle, provides insights into their preferences, and assists in identifying areas where they may need support. According to studies within the literature, the use of play therapy for children with autism has shown positive outcomes in fostering the development of social skills, reducing problematic behaviors, enhancing symbolic play skills, and promoting healthier communication with their families [1-3].

In the post-pandemic world, physical activities have been progressively losing their prominence. A reduction in physical activities can lead to significant health issues in later years. A sedentary lifestyle adversely affects children's body composition and overall health. For children with autism, this situation is exacerbated by additional challenges that impact their motor

skills, psychological and social development, and family life. A lack of physical activities can result in developmental regressions in these children [4,5].

Within the scope of the developed study, the objective was to equip children with autism with new skills through a combination of physical, cognitive, and emotional activities conducted in both classroom and home settings. In 2023, a game was specifically designed for children with autism to enhance their understanding of emotions and environmental awareness [9-13]. The game includes activities such as matching coins, identifying characters, and guessing emotions, each with levels tailored to different age groups. Similarly, in 2016, a cooperative puzzle game designed for two players was developed to encourage cooperation between children [14]. In 2019, another similar game, named "Altriras," was created [15-18]. All these studies targeted cognitive and emotional activities for children with autism.

In the newly developed study, games focusing on physical activities were designed and projected onto a board via a projector in the form of animations. These animations include virtual characters created by scanning the children, along with environmental elements and activities. The children complete the activities based on their movements. Unlike other similar studies, this study emphasizes physical activity and aims to enhance the social skills of children with autism.

Since other studies do not require physical movement, they often promote inactivity and contribute to a sedentary lifestyle. Therefore, expert opinions were sought for the development of this study. The dance activities and movement-based games, such as jumping and squatting, were approved by experts as effective in increasing physical activities for children. The developed study is enriched with various audio and visual elements to boost children's motivation for participation. Its interactive nature helps sustain children's attention for longer periods, making their developmental process more effective.

B. Aims and Objectives

Children often gain their experiences through play. When play therapy is utilized for children with autism, it has been observed to improve their communication skills, reduce behavioral problems, and support the development of their social abilities [9-13]. Play serves as a vital tool for children to

prepare for adulthood and to foster their holistic development. It has been highlighted that play contributes positively to both social and emotional development [14]. This proposed study aims to focus on the intrinsic potential and abilities of children with autism, supporting them not only in their current developmental stage but also in their journey towards becoming independent individuals in the future.

The study incorporates games specifically designed to facilitate the socialization of children with autism within their classrooms or home environments. Through these interactive games, children are provided with opportunities to enhance their communication skills without being isolated from their social community. These games are intended to strengthen their ability to express themselves and build essential friendship skills. Additionally, multiplayer games aim to help children with autism develop the ability to cope with emotional and behavioral challenges [15-18].

The activities included in the developed study encompass a variety of engaging and educational games: a matching game to enhance cognitive skills, an emotion-guessing game to improve emotional intelligence, a game to mimic movements on the screen to develop motor skills, a turn-based racing game for two players to foster cooperative play, and a dance activity that can involve all participants to promote physical coordination and social interaction. The visual elements used in these activities are carefully selected to maintain the child's attention and engagement with the game. These activities are carried out through a virtual character that operates synchronously with the child's physical activity, ensuring an immersive and interactive experience.

II. MATERIALS AND METHODS

The hardware used in this game software, which is developed to support the skill development of children with autism, includes the following components:

- **Computer:** The primary device on which the game software operates. It processes all the data and runs the interactive programs essential for the activities.
- **Projector:** Utilized to project the game onto a whiteboard in classroom settings, creating a large and engaging visual display for children.
- **Camera:** Captures user images and movements, serving as the main input device for the system. The camera's real-time feedback is crucial for the interactive nature of the games.

A. Methods

In the enhanced study, the first step involves scanning users with the help of a camera. Using the captured images, virtual characters are created through image processing techniques. These characters are presented to the users via a projector in classroom settings or through a computer monitor in home environments. Activities are conducted and controlled based on the users' physical movements. Upon the completion of these activities, children are rewarded, and data is recorded.

The software language and tools used in the preparation of these stages are as follows: C#, Unity, JSON, Blender, Adobe Photoshop, and Google MediaPipe. During this process, the developmental progress of children with autism is documented in the form of data. The study is supported by artificial intelligence, image processing algorithms, and geometric algorithms.

Image scanning and the processing of captured images are implemented using the C# programming language.

1) **Flowchart:** The flowchart followed within the scope of the enhanced study is presented in *Figure-1*.

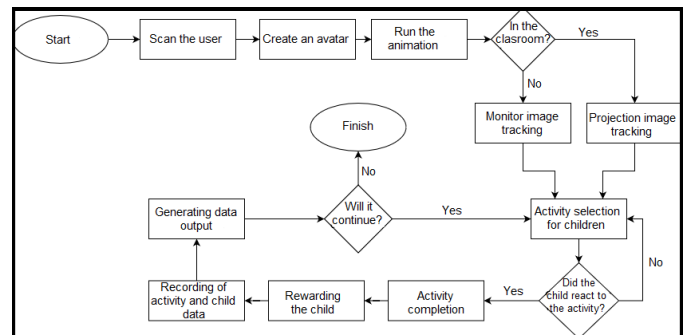


Figure-1. The General Structure of the System

2) **Visual Block Diagram:** The block diagram enriched with visuals describing the developed study is presented in *Figure-2*.

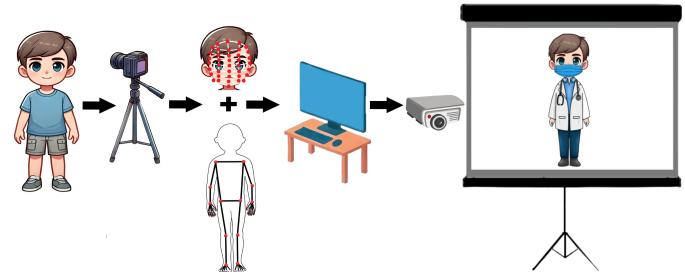


Figure-2. Detection of the Face and Body of the Individual in Front of the Camera

Figure-2 illustrates the stage where an individual steps in front of a camera to initiate the face and body detection process. In the first step, the individual positions themselves in front of the camera, and the system performs face and body detection. At this stage, the primary features of the face are identified, and a digital face map is created. Simultaneously, the system detects the joint points of the body, generating a digital body map. This process enables the analysis of facial expressions and gestures while marking the joint points to simulate body movements in alignment with real-life actions. The face detection process is particularly used to monitor children's responses and emotional states during activities. The body detection process, on the other hand, ensures that children perform the necessary movements to progress through games. Face and body detection data are transferred to a

3) **Image Processing with Google MediaPipe:** In this study, the Google MediaPipe library was used. MediaPipe employs an image processing and machine learning-based approach to perform functions such as full-body motion capture and finger tracking using a single camera. With multi-platform support and high accuracy, MediaPipe provides a more general and functional solution compared to other approaches.

The image processing task is performed in five key steps:

- Once user images are processed, they are transformed into virtual models. These virtual models are controlled in real-time, referencing the users' actual movements. An example visual of the system is presented in *Figure-3*.

A photograph of a young girl with dark hair, wearing a light-colored shirt. A white line with blue circular markers is overlaid on her face, tracking her facial features. The markers are positioned at the corners of her eyes, the bridge of her nose, the tip of her nose, the corners of her mouth, and a point on her chin. A horizontal line with two markers is also visible below her chin, likely representing the jawline or a reference line. The background is a plain, light-colored wall.

5) **Data Collection:** At the end of the activities, data is recorded to monitor the development of children with autism and to use in future research. These data help us understand how children perform during the activities and enable tracking of individual progress. The data is stored in JSON file format. The content of the data includes the date and duration of each activity, the type of activity, the child's preferences and interests, and information about which activity the child selected.

This study aims to contribute to the development of physical, cognitive, and social skills in children with Autism Spectrum Disorder (ASD) through a game-based, technology-supported application. The activities designed within the scope of the study have demonstrated positive effects on motor coordination, attention, and socialization skills, tailored to the specific needs of children.

Findings and Contributions

Contribution to Education and Technology: The study serves as a model for the effective use of technology in the education of individuals with autism and has become a significant reference source in this field. Furthermore, it has provided new data demonstrating the impact of game-based approaches in special education.

Societal Impact: The developed application offers a low-cost and accessible solution that can be implemented in diverse contexts such as preschools and home settings. It supports the integration of individuals with autism into society.

Future Directions

The results of this study indicate that game-based approaches are an effective tool for the education and rehabilitation of individuals with ASD. However, broader studies encompassing different age groups and larger populations are necessary. Recommendations for future research include:

- Evaluating the long-term effects of the developed system on a larger group of individuals,
- Customizing activities based on varying levels of ASD symptoms,
- Expanding the usability of the application by integrating it with different devices and platforms,
- Developing a multilingual interface and culturally sensitive content to provide a global solution,
- Implementing AI-powered personalization algorithms to better address individual needs.

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